

ActInf Textbook Group ~ Cohort 5 ~ Session 22 (Chapter 10, Part 2) ~ 4/8/2

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all right welcome back call 5 we're in
our last
recorded meeting Andrew and all of us
will be talking about chapter 10 but
also kind of probably anything
goes
and yeah let's go there Andrew however
you want to proceed yep sure yeah I'll
just give
a quick kind of summary of this chap I
cannot give an entire holistic summary
of this chapter because as we know it
longest chapter in the textbook um and
one of its key themes is to describe
active inference is a basically a kind
of complete solution to sentient
organisms adaptive problems which
attempts to provide a unified
perspective on action selection
perception attention emotion regulation
of right of other cognitive
mechanisms um so it's it's it attempts
to give us I think a kind of like let's
round out the textbook here let's let's
let's go over in section
10.2 all of the previous chapters up to
this point give a quick summary like
kind of in a per paragraph approach for
each chapter reminding us of anything
from marov blankets to neurobiological
sub substans message
passing uh building models and so on um
section 10.3 quickly mentions this uh
the famous uh philosopher Daniel dennet
his suggestion to model the whole iguana

uh was the was the quote and that's more or less what active inference attempts to do by using free energy minimization is a sort of first principle by which we can derive implications for a variety of cognitive

processes um it takes a kind of inactive integrative approach to to behavior to preference realization self- evidencing self-preservation and autop

poesis um it does not separate entirely action and perception as is done in uh or has been done at least in different um

neuropsychological uh

approaches and uh yeah it just attempts to yeah just take a very integrated view don't view action and perception as totally separate don't view Memory and attention is completely separate

planning prospection retrospection all these things are are tied together by way of a free energy minimization and then from there for anyone who's you know more more focused upon sort of the the neural and neurobiological substrates and

underpinnings neural signatures getting into neuroimaging Data then it might just be the case that we're looking at different Pathways uh for each of these things but nonetheless they're they're sort of of a piece um and before I I

wrap up this this summary I mentioned I I noticed at the end of section

10.3 they mentioned um uh reference this this great

neuroscientist uh up at NYU Yori busak

who's kind of famous for um back around 2006 I believe 2004 2006 um talked to began talking about this kind of inside out approach to studying the the brain and how the cognitive and psychological Sciences have sort of inherited all of these um Jamesy and William Jamesian uh categories in in how they understand the the mind the the mind brain connection and so on so these were even these very words that we're using uh memory and planning and emotion uh you can just flip open like one of James's like key books that's been used as a foundation in many Psychology and Neuroscience courses and just yeah look at its table of contents and we have so many of these these words that that are continuing to be used in a very sometimes very free floating kind of way uh in in current research um what does memory mean uh what what is attention exactly and uh yeah so it's it's not to say that all these Jamesian categories should be you know eliminated uh from from The Sciences or you know reject did or even uh supposed to have you know zero value they have plenty of value but uh it's just it's an interesting take to try and use what we know now about neurobiology about computation mathematics and yeah just maybe it's time for the the cognitive sciences and others to kind of update some of their their taxonomies and terminology uh that they use um for this

kind of work and and just consider like more integrative perspectives and for all these things that are treated separately so anyway that's kind of my two cents on that reference they make and

um so from there we get um you know these kind of mini subsections uh that relate active inference to just a broad variety of other fields um its relationships its different is um because they're able to find kind of positive relationships between active inference and these other fields it kind of opens the door for dialogue uh for control theory predictive processing um action control cybernetics idiom motor theory it's it's I think it's it's kind of it's it's doing exactly what it purports to do which is to find an integrated perspective um then that said it distinguishes itself um active inference is becoming its own kind of approach framework uh body of knowledge and so yeah it's uh it's just kind of placing itself in the sort of epistemic field that we we find ourselves within and um relationships with utility decision making reinforcement learning so many people I think come to active inference in part because they want to study those things whenever they're not coming from a more neuroscientific background and it's like well utility is kind of implicated in active inference but but it's more than that we have

epistemic value and we have balancing between epistemic and pragmatic value rather than treating them as their own separate cost functions or ignoring one of them or otherwise and finally the the chapter points at sort of these other fields in open directions um the idea of modeling social and cultural Dynamics is a kind of broader scale application of active inference um you could even presume that that goes down to the micro level that is we have human beings individuals with their own generative models just interacting in a kind of multi-agent environment and then from there checking to see what are the emergent behaviors whenever they interact and so what kind of kind of social dynamics happen what kind of cultural creations and Productions they they come up with um things like that so um I think I'll leave it there for now there's just too much to go over I won't give like a point by Point like here's the difference between active inference and um you know reinforcement learning right away or something but um that's about good stopping point I think so if anyone has any you know points of interests they want to talk about any kind of questions something on their mind U please feel free to go for it okay I have a have a more of a it's yeah try to explain it in simple words like I've you seen this use of this word epistemic a lot and it can make multiple things what does the word epistemic mean in this in

the context of active

inference

yeah sure I'll I'll I'll give that a

shot um so previously in the textbook at

times we would see you know we've seen

different like kind of equations that

break down computation of variational

and expected free

energies and so we've seen how it can be

broken down into word like certain terms

that that have been given a name so we

have like Risk ambiguity

um complexity accuracy so so two of

those terms that can be broken down into

are

pragmatic um value and epistemic value

and uh those are those are pretty

essential they relate to uh a classic

problem in the the social sciences

called the the explore exploit tradeoff

um but in simpler terms pragmatic value

as opposed to EP epistemic is something

like an agent just going for realizing

their goals directly like you're hungry

you want some food you go for the food

um you know realize that preference to

to you know diminish your hunger keep

your

homeostasis pretty aligned and balanced

um but epistemic value meanwhile would

be acting in order to get more

information

in order to realize your goal so again

you're hungry you want some food well

where can I get food like maybe it's

contextual where can I get food right

now um I you can use the internet search

up what places are open uh what what

places are delivering um how much money
do I have uh currently and what's my
budget so it would basically just be
kind of instrumentally you searching
seeking for information that can help
you to realize the goal so often times
Whenever there are like behavioral
experiments like a kind of two Arm
Bandit problem for example where an
agent can uh they can go
to uh they have multiple actions
available to them they can go uh
immediately to uh a reward like pulling
the slot machine at the the end of the
task or
prior to that they can go to a q
cuq that will inform them uh Which slot
machine to pull which one has the higher
probability of giving them a payoff and
so the agent um you know if the if the
agent doesn't care about the epistemic
value it could derive from the queue uh
giving it the information of how to
better realize its goal it'll just skip
the Queue go straight for for pragmatic
value pull the slot machine see if it
wins um so I I I hope that that was
helpful but I yeah I think that the
simplest terms are epistemic is
something like the knowing it's the it's
the information that'll help you get to
your goal pragmatic action and value are
derived from from actually going
directly at the goal so you collect
enough info you know what to do after
that you go for the pr Matic
value yeah a little bit of just other
broader pieces here

epistem in contrast with belief doca

like a

Creed also as distinguishing the the

knowledge from the Creed from the

technique which is the basis of

Technology um and then another helpful

piece like uh Andrew mentioned the

epistemic pragmatic value the epistemic

value is like known

unknowns in the prospective setting

because it's a value ascertainment about

a given policy how much I expect to

learn from that policy so of course if

something you didn't expect to learn

happened it couldn't have been

considered at the moment so that would

be an epistemic event like in the

broader sense of just epistemic meaning

learning or knowledge but when we're

talking about expect Ed free energy

equation

2.6 epistemic value is specifically

about expected Information Gain which

has to do with like how certain or

uncertain the agent is in the

moment and then the epistemic chaining

demo um which is just like one step it's

basically One Step more detail than in

the

textbook but um you see the mouse

navigate to The

Q but this just kind of begets other

questions like how does it come to

believe with with full Precision that

the Q is informative so this isn't the

whole iguana but this kind of isolates a

situation where either a risky pragmatic

decision can be approached immediately

or a little bit more of a circuitous path
can be taken that results in Information
Gain results in observations that are
expected to reduce uncertainty about
where the preferred observations
are and then depending on how all the
parameters are set you could have an
agent that just still goes and risk it
risks it for the uncertain reward
sooner can we can we actually phrase it
like uh epistemic uh value and epistemic
capacity is like the potential of the
possible things which can be done and
the pragmatic things the actual thing
that we do so it's like EP epistemic
values are like uh as a ground on which
possible things can happen and pragmatic
thing is the actual thing that you
do like conversationally that might make
sense but when you're talking when you
say things that you could do that's
basically affordances
and affordances then can be evaluated in
terms of their epistemic and pragmatic
value but it isn't that something that
you can do is related to The Information
Gain of that
action but I mean conversationally it
makes sense
like the broader epistemic concept the
agent has to know what it could do yeah
it has to be known what could be done so
it's not that other terms are expunged
of
epistemic components it's just that
epistemic value is being used in a
pretty narrow way in equation
2.6 in the expected Information Gain

that does highlight something very important but you could do kind of an epistemic inquiry into how does one know that they know and so on sorry I just uh could you just clearly differenti between this with affordances which you said I think what I saying is like so affordances is the possible things which a person can do at a particular moment of time right exactly yeah and policies like the the various policies you can do but affordance is like at that specific point of time what the agent does that's basically the affordances so affordance is very context sensitive yeah affordances are actions that can be taken in a moment policies are affordances over the time Horizon of planning so if the time Horizon of planning is just one then the policy space is the affordance space if you're talking about a time Horizon of two the policy space P_i the list or vector P_i that sums to one if you're talking about a time Horizon of two is going to be the length of affordances squared so like up down left right in the moment affordances and then planning Horizon of two so the policy space of which consideration is happening in would be 16 and then each of those 16 is going to have a prior so How likely is it a priori and then if it gets updated by expected free energy it's going to get updated according to

the Information Gain and the pragmatic value of each of those policies where did you get the number 16 like four into four four squared oh okay yeah and that's the kind of just like it's like a chess algorithm like the first pass approach is just to expand your policy space exponentially but then if you have more than a couple of affordances at each time step or you go more than a couple of time steps deep the space of the policy Vector becomes huge and then that necessitates all these different approaches to dealing with large policy spaces ranging from like pruning of policy trees to setting up the problem a little bit differently so that the branching is less exponential so here's the narrow epistemic value concept just KL Divergence fancy e big expectation of KL Divergence between r q variational approximation of hidden States Through Time conditioned upon in both besides the action everything is conditioned upon the action and the only difference is why Tilda so for that action how much will those observations coming in update my beliefs about hidden States through time if the observations are expected to be moot then the Information Gain is low

because it's like it might as well not be here if you expect there to be an observation coming in that's super informative then this is going to be a large

Divergence and that's kind of perfectly partitioned away from whether those observations coming in are aligned with what you expect and prefer to

see like pulling a slot machine like I mean there's there's a lot of ways to go with this but like you can imagine there's two agents they're both pulling a slot machine one of them is doing it picking that policy because they're seeking the information that the slot machine provides next another one is doing it because they believe that it's going to come up in a favorable setting for

them so that's kind of something people people are exploring with a Precision Psychiatry would be okay this person's engaged in this repetitive behavior is it because it's simply a habit and they don't prefer to do it is this happening because they actually prefer the outcome that it

reaches is this happening because they prefer an outcome and they've misunderstood the consequences of the action so they think that that action is going to reach their desired outcome but those from the outside look exactly the

same hence designing like experiments and cognitive models that could differentiate why those are

different I have one question could you
please go back to that last equation
which is
showing yeah over here actually I don't
know what this what does the term mean
this Q the last one that that what is
the operator equal to and a triangle on
top any
idea defined as equal to
okay defined as equal by definition
two equal by definition
to like this is the joint distribution
so here's Q that's our variational
approximator that we
control this is a joint distribution of
 X and Y conditioned upon P_i so it's like
there's kind of an X and a Y field
and we're making a probability density
over the X and the Y field conditioned
upon what action we're
selecting and then that is defined
as the joint product of our beliefs
about hidden State given
policy multiplied by basically the a
matrix the real mapping between hidden
States and observations so here's the
here's our our big underlying joint
generative model of the true temperature
in the room and the thermometer
reading conditioned upon
policy and we're going to be able to
break that down into well what is the
temperature of the room given our
policy and then what is the thermometer
reading given the true temperature of
room because of the factorizability the
the fact that the X and the Y were
orthogonal

gives us a sparsity that we can leverage

to

decompose The Joint distribution into

these two simpler

distributions

e

that's quite a thread

there it's like I've forgotten more than

I'll ever know just like how did this

happen deny the meaning of

awareness that's

um I do recommend I cannot remember the

name of the paper but there's like this

classic Marvin Minsky paper where he's

describing his take on AI at the time

this must be decades ago at this point

and at the end he hints at the idea that

uh and some kind of artificial

intelligence would have to have a model

itself and I just remember skimming that

paper and looking at active inference

and this idea that one has to have their

own model the of of themselves in the

world in order to kind of maintain that

that Markov blanket maintain that

separation between self and others so

that you can actually maintain your own

homeostasis maintain that um division

between between model an

environment kind of rambling on yeah and

like like we're aware of making

decisions on top of what we're not aware

of we're able to talk about this

semantics of like moving our hand around

because there's this kind of

infrastructural component that we're not

generally aware

of they call it self- evidencing

right yeah well self- evidencing let's
see where it comes
up self- evidencing is provided as
something like
the analogy to reward maximizing which
puts reward maximization as the unified
imperative in active inference where we
have surprise minimization or surprise
bounding about one's generative model as
the kind of key component we can think
about as I say one existential
imperative self-
evidencing but solipsism
is kind of a um deadend strategy so
self- evidencing doesn't just mean
observing oneself because over long
enough timelines that ends up becoming
maladaptive so that's kind of like well
how could novelty or creativity or
Aesthetics be brought
in when the unifying principle that
underlies behavior is to glean evidence
for one's model as it
is this is funny also like in chapter 10
has to come back to the preface so this
is I guess written by
Carl it's not just about
reading the free energy principle
equates existence with self-
evidencing can I tell a joke like I
don't know it's a joke or something they
they said that like all these people
this they're just worried about it the
mind will just shut off you know like
they will not do any self- evidencing
and then uh no nobody wants do that
something that is like Mo like like the
greatest fear among many people is that

okay they just don't want them they're
just worried about that the mind will
shut off one day and there'll be no
self-evidencing and everything will go
at us you know all this kind of
stuff h

sorry my level of jokes are with my kids
like the way I speak with my kids that
is the level of jokes I I don't know
better ones that's the sweet spot for
humor though it's like three to six
years yeah I mean it's it's
a strongly reinforced

theme that simple on through complex
systems are self-
evidencing to Garner sensory data
consistent

with an internal

model so it's like in the reward World
we'd have some temperature distribution
and then we have a second layer
Distribution on top of temperature how
rewarding those temperatures are and
then we pursue rewarding temperatures
like body temperature in the self-
evidencing world we have our basal
distribution of

temperature and then it's just explained
as action is taken to continually sample
from that distribution of preferred
temperatures whereas reward you need to
interpose with like another layer in
between the temperature and the
action so you're you have temperature
and then reward layer and then you're
taking actions based upon reward of
different actions here we have the basal
temperature layer and then we're taking

actions directly on that

distribution

um can I can I give a perspective of

self- evidencing as it relates to

meditation

yeah yeah okay I'm I'm not sure if I'm

in the right track like there will be

more learnings but self- evidencing like

it's actually people self-evidence is

there is a there's actually opposing

force between self- evidencing and

meditation basically because self-

evidencing is you want

to that's how I'm thinking like we know

ourselves by the evidences which are

provided from outside that is we are

inferring our evidences from the

knowledge from outside and um meditation

is trying to see what is really inside

so and and then there is one more this

concept marvelous concept of blankets

right so it's conditional Independence

you if the more you know about the

outside world if you know the blanket

States doesn't mean that you know more

about the inside world and that's like a

that's that's like I I I have just

fallen in love with this fundamental

concept like even if you know so much

about the external world but if the

blanket states are just this much then

knowing more about the external world

doesn't know more about yourself so

self- evidencing has its limits and uh

yeah so there is um there is some kind

a dynamic interplace so but but what I

think is this this this this

formulations and this concepts of active

influence can let us know so much about the mind and how it works uh but there will be there I don't know much but like there will be challenges when we explore the Realms beyond the mind it's the Consciousness so I don't want to talk about that because people can say and they can have lot of theories but it may not necessarily be right because this conditional Bound by this Marco BL blanket presents a very fundamental um that's how that's like I don't know if I phrased the appropriately but like that is like this these two pillars are what I'm basically thinking and how it relates to meditation and these Concepts yeah yeah it makes me think of U I don't know if this directly relates to what you're describing but this whole you know just because you might know a lot about the outside doesn't necessarily mean you know a lot about the inside and it makes me think about you know similar to the rest of of kind of current research and n neoscience uh active inference also splits up uh when describing different sensory modalities as you know we don't just have like one kind we have proprioception and exteroception and interoception and all three of them relate to inference and kind of having to you know update beliefs change your attention um and and and kind of figure out what to do and so in Tarot SE means like reading your own body and reading your own feelings even this this chapter 10 here relates um there's like a

section entire section that relates to
like emotion um in in trying to describe
that right like you have to actually
infer how you're feeling you don't just
automatically you know perfectly
precisely know and then suddenly an
emotion isn't so much like some kind of
innate perfectly inbuilt thing that we
just automatically like understand like
oh I'm sad it's like no you have to
infer that you have to like read what's
going on inside of you just as much as
you have to read what's going on outside
of you so suddenly your emotion is yeah
it's like a hypothesis of like oh based
on this feeling and this and this and
this other sensory information that I'm
getting from within me like how am I
feeling how am I doing let's put a
conceptual you know word on that name of
some emotion um yeah interesting uh
Kristoff you have a your hand
up uh yeah I just posted a link uh to a
talk by uh Shamil uh chandaria I think
he's affiliated with University of
Oxford and a bunch of other places and
it's a um Nono all Awakening yeah
computational neurom phenomenological
account uh which is basically entirely
based on the free energy principle
active in and it's an it's an incredible
talk it's the best summary of um
essentially various aspects of spiritual
practice from the perspective of
computational Neuroscience that they
have ever encountered and it's is really
really he goes really really deep um I
can tell you like from from my

perspective um understanding the free
energy principle um has been
enlightening to to say the least like in
in the in in the for me it's actually in
the literal sense there's a concept in
Buddhism which I believe is uh
Liberation by view um where um you know
a or or a practitioner
um receives a certain level of of of
Liberation by essentially understanding
um um certain truths about the nature of
reality without necessarily realizing um
uh for themselves a a certain meditative
attainments which are
um available to practitioners in the
path of insight so if you practice
mindfulness and concentration there is a
whole ladder of
um uh insights that you you essentially
climb you know the the the four janas
the the for the formless Realms and
finally the attainment of arand ship and
I find it that when I look at the
teachings of the Buddha particularly um
you know say reading the nikayas the
maim Manaya the middle length discourses
or s n the connected discourses and
thinking about them in terms of um free
energy principle and active inference um
the teachings of the Buddha are
incredibly clear like he is giving you
recognition models he's giving you means
of building your own um uh generative
models and it has a lot to do with
essentially making the pH phenomena that
happen in the mind opaque because
normally they're transparent to us we
don't know how they work we don't know

how the reality is uh constructed by the
mind we are not aware um of causes and
effects that give rise to certain
thoughts and feelings and and
perceptions and and mental formations
and concoctions and all these other
things and Buddha is giving you
essentially a
vocabulary so that you can um later on
label these things as they occur so he's
giving you a recognition model you can
say like ah this is a this is a
perception this is a mental formation
this is a feeling and feeling is born of
contact and contact is you know between
the internal sense basis and the
external sense basis and so on so forth
and then it that's for the purpose of
effectively building a generative model
um once you have the generative model
then you know you you get you become you
can become more introspective about your
own experience and
um the the goal of that is that your
experience becomes increasingly opaque
uh you see more aspects of it and so on
so forth and uh at the end of the path
you basically have complete insight into
how the reality is constructed by the
mind and that's why this phrase always
um recurs in in the Buddhist text that
you know um um the biku who completed
the path understands the the the the
birth is destroyed the holy life has
been lived what what what had to be done
has been done and there is no more
coming into any state of being it's not
a it's not a matter of you know question

like once once you get there you know
you know you're there so you know like
it's um I I think I I think there is a
the you know going back to this earlier
statement about the necessity of
updating the terminology and
taxonomy once you start looking at at
Buddhist teachings through this
perspective and if you look at this for
this modern perspective of computational
Neuroscience it's it's an Incredible
manual and it's so straightforward there
is no there is no um there's no doubt
about what you need to do and how how to
do it so I highly recommend um this talk
and uh you know it's a it's it's it's a
very fruitful area of
inquiry awesome

Francis thanks uh firstly uh I'd endorse
that uh thanks for um poting out to me
Kristoff uh

I uh watched it twice uh I'm sure I've
still got a hell of a lot more to get
out of it um it's

a really interesting

um secondly just kind of a related um
things to a very simple but it struck me
as interesting point uh which relates to
the whole idea of making the
transparent uh

opaque um and the fact that by making it
a p you can see it

which of course sounds a little

paradoxical um I've been thinking about

um mindfulness and kind of phenomenology

because it's really hard phenomenology

you know what is it that we're feeling

this stuff that we're not normally aware

of and of course a common mindfulness technique is to label your thoughts

and of course we normally think of that is oh we're labeling them because we want to then let them go but of course by labeling them you make them opaque and you could get a very good stream of

Consciousness

um uh by just speaking these labels loud um by

pursuing

the um St of mindfulness and the empty mind you actually become aware of what's in the mind and then um the third thing just kind of going back to this homeostasis

interception um and something even simpler uh which is when I was first started studying to be counselor I was introduced

to uh

C's uh learning Loop

uh Kolb and that has four axes which are it's kind of used in an educational context and uh they have these kind of rather long labels but then there's another variation where the four axes have one word labels and the labels would

be uh action feeling thought planning

and that

I've

explained that kind of gives us a model of how we process our feelings um and I explain it to people

in a counseling context why is it count
counselors are always asking you how
this makes you feel how you feel about
this
but you know we've got this engine this
cognitive engine up
here but how do we know what what's
steering it what's the
point our feelings are where the rubber
hits the
road the road being our
well-being if we don't
know how we're
feeling we don't
know what to do with our
cognition
um and as you get further away from
basic biological homeostasis homeostatic
demands the biological
needs uh it gets harder and harder
to engage with are more complex in a
leads demands between say status and
connection uh should I pursue a job or
should I make time for being with my
partner uh and especially when you enter
this uh existential
anomaly all these brain power cognitive
power helps to increase our survival
that's what it's there
for but what happens when we get so much
cognitive power that we realize that
none of
the all all our paths lead to nons
survival in the end and into the
existential
challenge little dump mind dump there
can I just add something about labeling
um popped in my mind um as you were

speaking uh so that's there's a that's
mahasa tradition um uh is essentially to
just practice labeling all day long you
just label everything you do but um I
found I found I find this uh really
there's this really interesting paper um
from the deep Learning Community called
your classifier
is secretly an energy based model and
you should treat it as such and um I
think it's by Will
Grall grad wolf I'm not sure what the C
name is but anyway so they show that if
you train um you know standard
classifier train on across entropy
objective like typical you know amidst
classifier neural network there's a way
of interpreting that as a energy based
model in such a way that you get
effectively the kind of unnormalized
probabilities that are required to
compute free energy and you effectively
get um uh a generative model for
free so this idea you know of labeling
by m and and other traditions in
Buddhism that all you have to do is
labeling
it's actually uh you know if you squint
your eyes at this paper and you say like
you know learning is kind of universal
and no matter how it's implemented the
the same sort of effects take place you
are B that that that is exactly the way
to build a generative
model um which I find I find it I I
think this this formal equivalence
between uh classification uh uh you know
the recognition and generative models

through energy based models specifically
which is just such a beautiful enrichment
formalism especially and and so
connected to to active inference um
that's going to be um is going to be
important so I I think this is a very uh
very interesting uh research Avenue of
how to connect active inference with um
this deep learning world and in some
sense you know like the because active
inference and free energy are so
Universal phenomena we know through this
argument that because they're apply to
anything that exists mathematic a prior
then these these systems must be
building models in the basian sense from
a certain perspective but it seems that
that equivalent is a lot deeper and much
more direct than just um you know this
kind of this is a mathematical principle
that has to be true kind of argument
it's actually can be shown how uh how
that happens so yeah uh fascinating
paper I highly recommend
it
yeah super cool it's fascinating to see
just like with this the integrative
approach of this last chapter how many
possible topics we could land on during
this conversation from from Deep
learning mathematical
formalisms uh the excesses of the the
human cognitive mind um
yeah it's it's it's really interesting I
personally just I find this kind of
first principles approach of act active
inference to just open doors to many
different

dialogues um and Francis I was finding that to be very interesting I had like a a kind of naive take on uh the idea of like what what happens whenever you start getting away from like a parent homeo static needs um from an actm perspective which is you know from a a modeling perspective we're looking at like an agent has their preferences uh which are really their their expectations uh about preferences and and so there are things that you expect more to occur um and those those are you know your your wants your needs and so on might be some kind of colloquial language we could assign to those uh prior or those preferences and like the idea might be that as a human being navigates the world going forward an increasingly more complex world you know they they start to learn that they need to do things Beyond simply oh I need to drink water to say sh thirst um like a variety of things navigating a kind of Legal Financial social occupational familial world that they're in and it's almost like there's this kind of expansion of uh of the state space to learn about and uh with that a kind of expansion of observations or preferences to be realized in order to to meet our wants and our our needs and and and so to bring it back to this discrepancy between like oh just keeping your heartbeat uh regulated keeping your your hydration levels reasonable like suddenly there're all kinds of things I

want a nice house and I want a I want a
nice car or or conversely I I want to
have a very energy
efficient uh green uh lifestyle uh or
some such thing and and as it becomes
more complex it's like it just makes me
think of uh this classic Rolling Stones
song uh You Can't Always Get What You
Want

um and you know sometimes you can get
what you need um it also brings me back
to Buddhism and a lot of commentary upon
the role of of desire and its relation
with with suffering um just kind of you
know Li desire can ultimately be be
Limitless it can be transient uh it can
be something that sort of leads you
astray from the path um can be all kinds
of things and yeah I don't know I I
don't know if it's about kind of raining
in our our desires every once in a while
and really evaluating like what is
important here uh what do we really need
to do um and and how can we how can we
have our basic needs met then from there
I look at like look to neuroscience and
uh folks like um there's someone named
phel uh Sarah Felman Barrett she's great
she's done a lot of work on U uh emotion
and affect and and she's at least a
couple of times had some collaboration
with Carl friston and other folks in
active
inference um and she says like sometimes
you just need some good sleep some
sometimes whenever there's a kind of
overwhelmingness a kind of you begin to
interpret what's going on with you is

like anxiety uh you know in the extremes
of kind of catastrophic thinking of
trying to do everything all at once you
can imagine an active inference agent
trying to just minimize a ton of free
energy all at one time and it cannot
find the right policy to to accomplish
that within five seconds as much as it
might wish that such a thing were
possible um yeah it's really just you
know get some good sleep can you do that
and you know get some get some good
nutrients in your body and maybe get
whatever kind of exercise is reasonable
to you if if only like a nice walk um
apparent that can heavily change the the
inferential process uh and maybe can
help with straightening out thinking um
I could ramble on forever but yeah I'm
just kind of uh wrapping up the the
meeting here some thoughts throwing them
all

around I have to leave in one minute
because I've got a client uh on the hour
can I just uh yes please very thank you
so much and there's so much more love to
say and to talk about uh and I'm really
sorry that we've come to our last
session yes for this little fractal
cycle but we will all be able to suffer
more looking at pdps in the
future I'll be there with
luck okay thanks guys well thank you so
much thanks everyone yeah thank you okay
next week um we'll we'll just see what
happens we can um fill out the feedback
form
anonymously look through some of these

many comments and also I think now that
we are two years after the the
publishing of this book March uh
22 we can start to think about you know
we have RX and fur we have all these
other activities how do we want to
continue with the par our textbook group
so think about it over the next week and
then we'll discuss more L later thanks
thanks Andrew thanks
everybody thanks everybody it's been
amazing
yay thanks folks